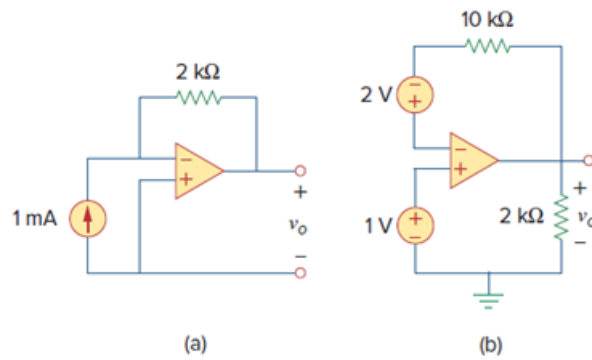


Problem

Obtain v_o for each of the op amp circuits in Fig. 5.47.

Figure 5.47:



Step-by-step solution

Step 1 of 5

(a)

Refer to Figure 5.47 (a) in the text book.

Draw the circuit diagram with node voltages.

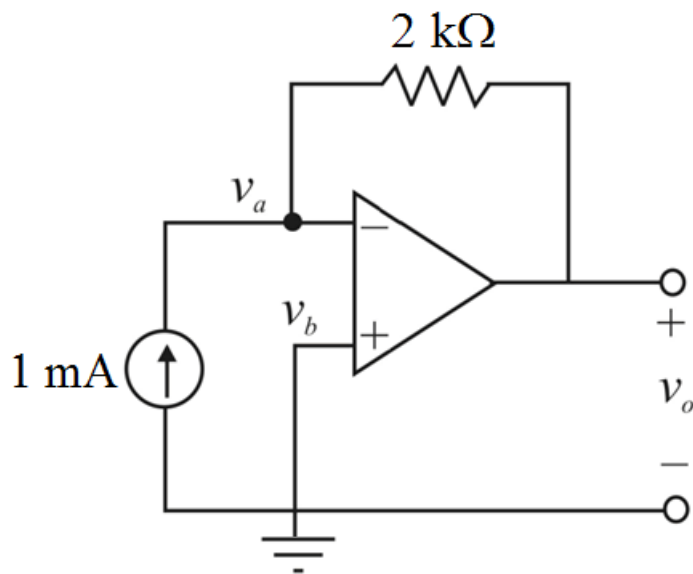


Figure 1

Step 2 of 5

Assume that the op-amp is ideal.

From Figure 1, the voltage at non-inverting terminal is zero.

$$v_b = 0 \text{ V}$$

From the virtual ground concept, the voltage at the inverting terminal is equals to the voltage at non-inverting terminal.

$$\begin{aligned} v_a &= v_b \\ &= 0 \text{ V} \end{aligned}$$

Step 3 of 5

Apply Kirchhoff's Current Law at non-inverting terminal in Figure 1.

$$\begin{aligned} (-1 \times 10^{-3}) + \frac{v_a - v_o}{2 \times 10^3} &= 0 \\ \frac{(-1 \times 10^{-3})(2 \times 10^3) + (v_a - v_o)}{2 \times 10^3} &= 0 \\ -2 + v_a - v_o &= 0 \end{aligned}$$

Substitute 0 V for v_a in the equation.

$$\begin{aligned} -2 + 0 - v_o &= 0 \\ v_o &= -2 \text{ V} \end{aligned}$$

Thus, the value of output voltage, v_o is -2 V.

Step 4 of 5

(b)

Refer to Figure 5.47 (b) in the textbook.

Draw the circuit diagram with node voltages.

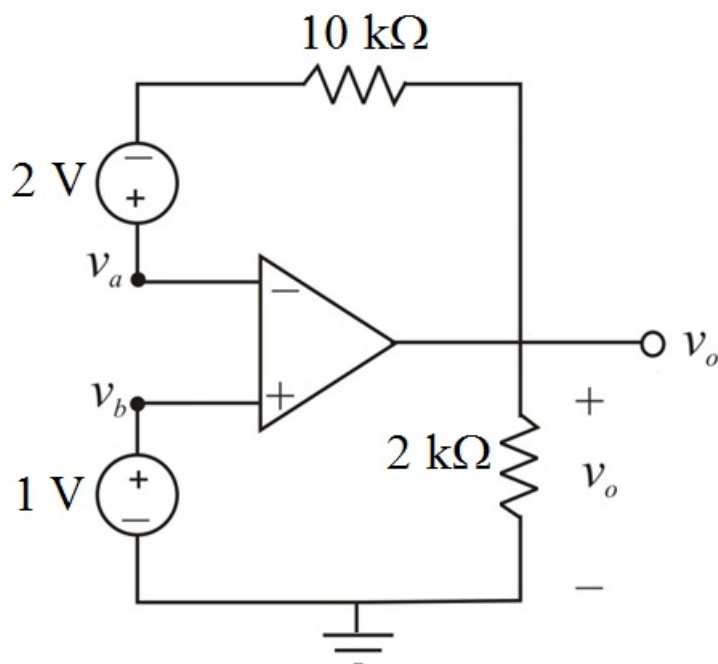


Figure 2

Step 5 of 5

Assume that the op-amp is ideal.

The voltage at inverting terminal is same as the voltage at non-inverting terminal.

$$v_a = v_b$$

From Figure 2, the voltage at non-inverting terminal is,

$$v_b = 1 \text{ V}$$

Therefore,

$$\begin{aligned} v_a &= v_b \\ &= 1 \text{ V} \end{aligned}$$

Apply the Kirchhoff's Current Law at node v_a .

$$\frac{v_a - 2 - v_o}{10 \times 10^3} = 0$$

$$v_a - 2 - v_o = 0$$

Substitute 1 V for v_a in the equation.

$$1 - 2 - v_o = 0$$

$$-1 - v_o = 0$$

$$v_o = -1 \text{ V}$$

Thus, the value of output voltage, v_o is $\boxed{-1 \text{ V}}$.